

pfSense Firewall Lab

1. What You Are Doing

In this lab, you will create a small virtual network with three machines:

- A firewall (pfSense) that controls traffic
- An attacker machine (Kali Linux)
- A victim machine (Ubuntu)

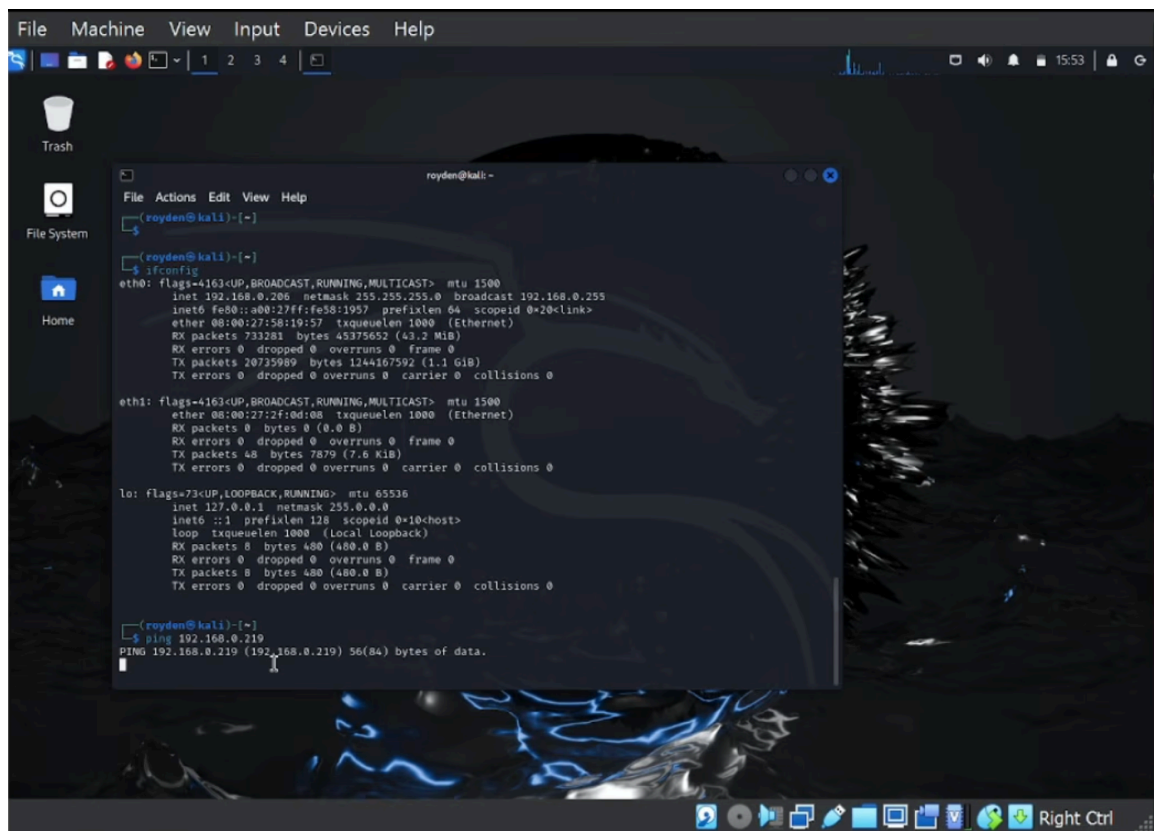
You will send a fake attack from the attacker and then block it using the firewall.

2. Basic Setup

Turn on all machines.

pfSense will automatically give the Ubuntu machine an IP address.

Make sure Kali and Ubuntu can communicate (ping test).



The screenshot shows a Kali Linux desktop environment with a terminal window open. The terminal displays the output of the `ifconfig` command, showing network details for `eth0`, `eth1`, and `lo`. Below the network configuration, a ping test is performed from the Kali machine to the Ubuntu machine (192.168.0.219), showing a successful connection with 56(84) bytes of data.

```
royden@kali:~$ ifconfig
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
    inet 192.168.0.200  netmask 255.255.255.0  broadcast 192.168.0.255
    inet6 fe80::a00:27:ff:fe58:1957  prefixlen 64  scopeid 0<link>
    ether 08:00:27:58:19:57  txqueuelen 1000  (Ethernet)
    RX packets 733281  bytes 45375652 (43.2 MiB)
    RX errors 0  dropped 0  overruns 0  frame 0
    TX packets 20735989  bytes 1244167592 (1.1 GiB)
    TX errors 0  dropped 0  overruns 0  carrier 0  collisions 0

eth1: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
    ether 08:00:27:2f:0d:08  txqueuelen 1000  (Ethernet)
    RX packets 0  bytes 0 (0.0 B)
    RX errors 0  dropped 0  overruns 0  frame 0
    TX packets 48  bytes 7879 (7.6 KiB)
    TX errors 0  dropped 0  overruns 0  carrier 0  collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING>  mtu 65536
    inet 127.0.0.1  netmask 255.0.0.0
    inet6 ::1  prefixlen 128  scopeid 0<host>
    loop txqueuelen 1000  (Local Loopback)
    RX packets 0  bytes 480 (480.0 B)
    RX errors 0  dropped 0  overruns 0  frame 0
    TX packets 8  bytes 480 (480.0 B)
    TX errors 0  dropped 0  overruns 0  carrier 0  collisions 0

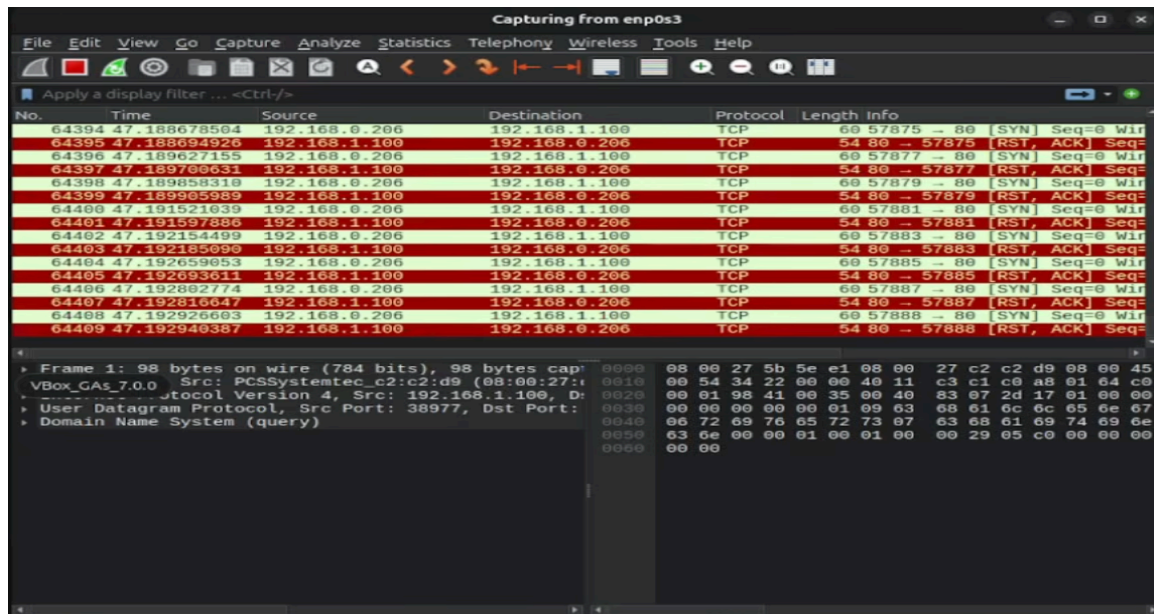
royden@kali:~$ ping 192.168.0.219
PING 192.168.0.219 (192.168.0.219) 56(84) bytes of data.
```

3. Simulate an Attack

On Ubuntu, open Wireshark to watch network traffic.

On Kali, run a command to send a large amount of traffic to Ubuntu.

You should see a spike in traffic on Ubuntu.



4. Block the Attack

Go to the pfSense web interface.

Create a rule to block traffic coming from the Kali machine.

Apply the rule

Rules (Drag to Change Order)											
☐	States	Protocol	Source	Port	Destination	Port	Gateway	Queue	Schedule	Description	Actions
☐		0/38.12 MiB	IPV4 TCP	192.168.0.206	*	192.168.1.100	*	*	none		

Continue the attack.

```
royden@kali: ~  
File Actions Edit View Help  
64 bytes from 192.168.1.100: icmp_seq=82 ttl=63 time=0.962 ms  
^C  
— 192.168.1.100 ping statistics —  
82 packets transmitted, 13 received, 84.1463% packet loss, time 82758ms  
rtt min/avg/max/mdev = 0.710/0.982/1.307/0.194 ms  
  
(royden@kali)-[~]  
$ sudo hping3 -S -p 80 --flood 192.168.1.100  
HPING 192.168.1.100 (eth0 192.168.1.100): 5 set, 40 headers + 0 data bytes  
hping in flood mode, no replies will be shown  
^C  
— 192.168.1.100 hping statistic —  
522773 packets transmitted, 0 packets received, 100% packet loss  
round-trip min/avg/max = 0.0/0.0/0.0 ms  
  
(royden@kali)-[~]  
$  
  
(royden@kali)-[~]  
$  
  
(royden@kali)-[~]  
$  
  
(royden@kali)-[~]  
$  
  
(royden@kali)-[~]  
$  
  
(royden@kali)-[~]  
$ sudo hping3 -S -p 80 --flood 192.168.1.100  
HPING 192.168.1.100 (eth0 192.168.1.100): 5 set, 40 headers + 0 data bytes  
hping in flood mode, no replies will be shown
```

The traffic should stop immediately.

```
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help  
Apply a display filter ... <Ctrl-/>  
No. Time Source Destination Protocol Length Info  
1 0.000000000 192.168.1.100 192.168.0.217 TCP 74 35368 → 8080 [SYN] Seq=0 Wi  
2 1.032966264 192.168.1.100 192.168.0.217 TCP 74 [TCP Retransmission] 35368  
3 2.037896576 192.168.1.100 192.168.0.217 TCP 74 [TCP Retransmission] 35368  
4 3.060949708 192.168.1.100 192.168.0.217 TCP 74 [TCP Retransmission] 35368  
5 4.085684788 192.168.1.100 192.168.0.217 TCP 74 [TCP Retransmission] 35368  
6 5.300916196 PCSSystemtec_c2:c2: PCSSystemtec_5b:5e: ARP 42 Who has 192.168.1.1? Tell 1  
7 5.301613298 PCSSystemtec_5b:5e: PCSSystemtec_c2:c2: ARP 60 192.168.1.1 is at 08:00:27:  
8 27.393327626 192.168.1.100 192.168.1.1 DHCP 329 DHCP Request - Transaction  
9 27.393807607 192.168.1.1 192.168.1.100 DHCP 342 DHCP ACK - Transaction  
  
Frame 1: 74 bytes on wire (592 bits), 74 bytes captured on interface eth0  
Ethernet II, Src: PCSSystemtec_c2:c2:d9 (08:00:27:00:00:00), Dst: 08:00:27:00:00:00  
Internet Protocol Version 4, Src: 192.168.1.100, Destination: 192.168.0.217  
Transmission Control Protocol, Src Port: 35368, Destination Port: 8080  
Sequence Number: 0, Window Size: 0, Length: 40, Options: SYN, Seq=0, Win=0, Len=0  
0000 08 00 27 5b 5e e1 08 00 27 c2 c2 d9 08 00 45  
0010 00 3c d6 1f 40 00 40 06 e1 0e c0 a8 01 64 c0  
0020 00 d9 8a 28 1f 90 6b 08 46 20 00 00 00 00 a0  
0030 fa f0 83 bc 00 00 02 04 05 b4 04 02 08 0a f2  
0040 29 97 00 00 00 00 01 03 03 07
```

5. What You Learned

- How a firewall monitors and controls traffic
- How attacks look on a network
- How to block unwanted traffic by creating rules